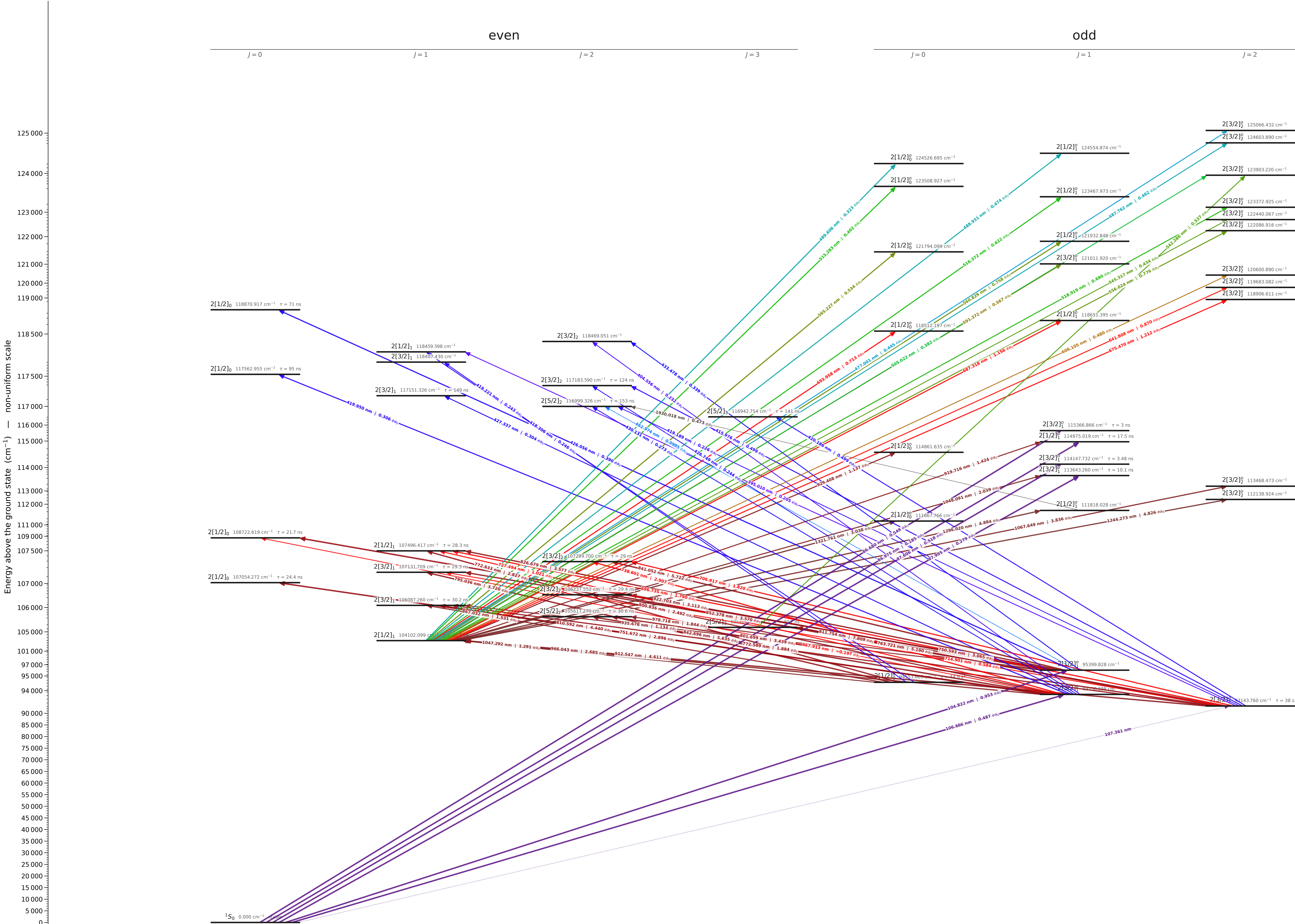
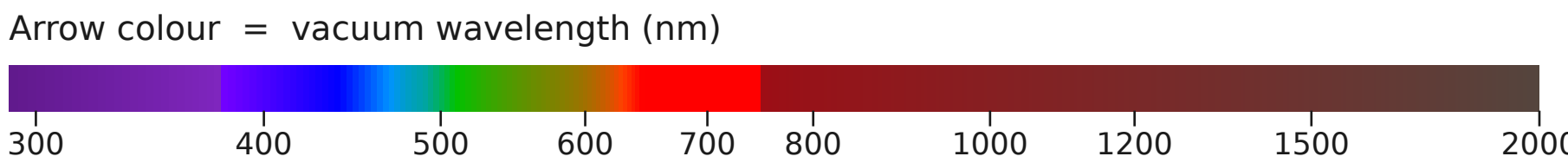


# Argon

Ar (Ar I). Strongest classified transitions below 2 μm (NIST): 51 levels, 75 lines.

Ar<sup>+</sup> ionisation limit 127109.84 cm<sup>-1</sup> ≈ 15.75961 eV (78.672 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element  $|\langle J \| e r \| J' \rangle|$  in ea<sub>0</sub>  
Matrix elements are measured or NIST values; \* = high-accuracy theory, = = model or semi-empirical calculation.  
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.  
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



## Key transitions

| Transition                                                          | λ vac (nm) | λ air (nm) | ν (THz)        | $ \langle J \  e r \  J' \rangle $ (ea <sub>0</sub> ) | A (s <sup>-1</sup> ) | Γ/2π up. (MHz) | branch (%) | Use                                                   |
|---------------------------------------------------------------------|------------|------------|----------------|-------------------------------------------------------|----------------------|----------------|------------|-------------------------------------------------------|
| <sup>1</sup> S <sub>0</sub> - 2[1/2] <sup>o</sup> <sub>1</sub>      | 104.8220   | -          | 2860.01500(10) | 0.953                                                 | 5.320e+08            | -              | -          | 104.8 nm resonance line                               |
| <sup>1</sup> S <sub>0</sub> - 2[3/2] <sup>o</sup> <sub>2</sub>      | 107.3609   | -          | 2792.3797      | -                                                     | 2.630e-02            | 4.188e-09      | 99.9       | 107.4 nm M2 decay of the metastable 3P2 state         |
| 2[3/2] <sup>o</sup> <sub>2</sub> - 2[3/2] <sup>o</sup> <sub>1</sub> | 763.7207   | 763.5105   | 392.5420       | 5.190                                                 | 2.450e+07            | 5.413          | 72         | 763.5 nm                                              |
| 2[3/2] <sup>o</sup> <sub>2</sub> - 2[3/2] <sup>o</sup> <sub>1</sub> | 800.8358   | 800.6156   | 374.3495       | 2.492                                                 | 4.900e+06            | 5.413          | 14.4       | 800.8 nm optical pumping 1s4 -> 2p6 -> metastable 1s5 |
| 2[3/2] <sup>o</sup> <sub>2</sub> - 2[5/2] <sup>o</sup> <sub>3</sub> | 811.7542   | 811.5311   | 369.3143       | 7.809                                                 | 3.300e+07            | 5.184          | 100        | 811.5 nm                                              |
| 2[1/2] <sup>o</sup> <sub>1</sub> - 2[3/2] <sup>o</sup> <sub>1</sub> | 922.7029   | 922.4498   | 324.9068       | 3.113                                                 | 5.000e+06            | 5.413          | 14.7       | 922.7 nm optical pumping 1s2 -> 2p6 -> metastable 1s5 |

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.  
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

## Isotope shifts

| Transition                                                          | λ vac (nm) | Isotopes | Shift (MHz) |
|---------------------------------------------------------------------|------------|----------|-------------|
| <sup>1</sup> S <sub>0</sub> - 2[1/2] <sup>o</sup> <sub>1</sub>      | 104.822    | 40-38    | 550(10)     |
| <sup>1</sup> S <sub>0</sub> - 2[1/2] <sup>o</sup> <sub>1</sub>      | 104.822    | 40-36    | 1036(7)     |
| 2[3/2] <sup>o</sup> <sub>2</sub> - 2[3/2] <sup>o</sup> <sub>1</sub> | 763.721    | 32-38    | -882(4)     |
| 2[3/2] <sup>o</sup> <sub>2</sub> - 2[3/2] <sup>o</sup> <sub>1</sub> | 763.721    | 33-38    | -703.2(21)  |
| 2[3/2] <sup>o</sup> <sub>2</sub> - 2[3/2] <sup>o</sup> <sub>1</sub> | 763.721    | 34-38    | -552.6(7)   |
| 2[3/2] <sup>o</sup> <sub>2</sub> - 2[3/2] <sup>o</sup> <sub>1</sub> | 763.721    | 35-38    | -394.4(27)  |
| 2[3/2] <sup>o</sup> <sub>2</sub> - 2[3/2] <sup>o</sup> <sub>1</sub> | 763.721    | 36-38    | -265.4(4)   |
| 2[3/2] <sup>o</sup> <sub>2</sub> - 2[3/2] <sup>o</sup> <sub>1</sub> | 763.721    | 37-38    | -124.2(11)  |
| 2[3/2] <sup>o</sup> <sub>2</sub> - 2[3/2] <sup>o</sup> <sub>1</sub> | 763.721    | 39-38    | 122(7)      |
| 2[3/2] <sup>o</sup> <sub>2</sub> - 2[3/2] <sup>o</sup> <sub>1</sub> | 763.721    | 40-38    | 229.5(4)    |
| 2[3/2] <sup>o</sup> <sub>2</sub> - 2[3/2] <sup>o</sup> <sub>1</sub> | 763.721    | 46-38    | 835.2(23)   |
| 2[3/2] <sup>o</sup> <sub>2</sub> - 2[3/2] <sup>o</sup> <sub>1</sub> | 800.836    | 36-40    | -640(10)    |
| 2[3/2] <sup>o</sup> <sub>2</sub> - 2[3/2] <sup>o</sup> <sub>1</sub> | 800.836    | 38-40    | -290(10)    |
| 2[3/2] <sup>o</sup> <sub>2</sub> - 2[3/2] <sup>o</sup> <sub>1</sub> | 800.836    | 39-40    | -128(16)    |
| 2[3/2] <sup>o</sup> <sub>2</sub> - 2[5/2] <sup>o</sup> <sub>3</sub> | 811.754    | 36-38    | -242.23(25) |
| 2[3/2] <sup>o</sup> <sub>2</sub> - 2[5/2] <sup>o</sup> <sub>3</sub> | 811.754    | 40-38    | 207.9(9)    |
| 2[3/2] <sup>o</sup> <sub>2</sub> - 2[5/2] <sup>o</sup> <sub>3</sub> | 811.754    | 39-38    | 113.9(4)    |
| 2[3/2] <sup>o</sup> <sub>2</sub> - 2[5/2] <sup>o</sup> <sub>3</sub> | 811.754    | 37-38    | -113.0(27)  |
| 2[3/2] <sup>o</sup> <sub>2</sub> - 2[5/2] <sup>o</sup> <sub>3</sub> | 811.754    | 36-40    | -450.1(9)   |
| 2[3/2] <sup>o</sup> <sub>2</sub> - 2[5/2] <sup>o</sup> <sub>3</sub> | 811.754    | 39-40    | -94.0(10)   |
| 2[1/2] <sup>o</sup> <sub>1</sub> - 2[3/2] <sup>o</sup> <sub>1</sub> | 922.703    | 36-40    | -960(10)    |
| 2[1/2] <sup>o</sup> <sub>1</sub> - 2[3/2] <sup>o</sup> <sub>1</sub> | 922.703    | 38-40    | -445(10)    |

Shift = ν(first isotope) - ν(second isotope).

## <sup>39</sup>Ar hyperfine structure (I = 7/2)

| Level                            | A (MHz)     | B (MHz)   | Shift of each F level from the centre of gravity (MHz)                                                         |
|----------------------------------|-------------|-----------|----------------------------------------------------------------------------------------------------------------|
| 2[3/2] <sup>o</sup> <sub>2</sub> | -285.36(13) | 117.8(12) | F=3/2: +2631.35 F=5/2: +1865.36 F=7/2: +822.42 F=9/2: -468.01 F=11/2: -1968.07                                 |
| 2[3/2] <sup>o</sup> <sub>2</sub> | -334.0(20)  | -24(3)    | F=5/2: +1490.14 F=7/2: +351.14 F=9/2: -1175.00                                                                 |
| 2[1/2] <sup>o</sup> <sub>1</sub> | -712.0(10)  | 84(6)     | F=5/2: +3249.00 F=7/2: +652.00 F=9/2: -2471.00                                                                 |
| 2[5/2] <sup>o</sup> <sub>3</sub> | -134.36(8)  | 113.1(14) | F=1/2: +1874.45 F=3/2: +1652.71 F=5/2: +1288.54 F=7/2: +790.00 F=9/2: +168.42 F=11/2: -561.68 F=13/2: -1382.51 |
| 2[3/2] <sup>o</sup> <sub>1</sub> | -163(31)    | -63(11)   | F=3/2: +1433.25 F=5/2: +1053.88 F=7/2: +507.00 F=9/2: -223.12 F=11/2: -1156.75                                 |

Measured A, B constants; sources in the data file.